

ORIGINAL ARTICLE

FLNC-Related Cardiomyopathies in an Admixed Population: Expanding the Genotype-Phenotype Spectrum and Clinical Correlates

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BACKGROUND: Pathogenic and likely pathogenic variants in filamin C (*FLNC*) cause a broad spectrum of cardiomyopathies. However, available genotype-phenotype data are derived predominantly from European and North American populations, with limited representation of admixed populations.

METHODS: We characterized carriers of pathogenic/likely pathogenic *FLNC* variants identified among 2654 unrelated probands undergoing exome sequencing for suspected inherited cardiomyopathy and their relatives identified through cascade screening. Clinical, electrocardiographic, imaging, genetic, and longitudinal data were analyzed according to cardiomyopathy phenotype and variant class. Genetic ancestry was evaluated. Associations with arrhythmic, heart failure, and composite outcomes were examined in exploratory analyses.

RESULTS: The cohort included 97 carriers (58 probands, 39 relatives), of whom 72 had cardiomyopathy and 25 were phenotype-negative at evaluation. Truncating variants predominated among patients with arrhythmogenic cardiomyopathy and dilated cardiomyopathy, whereas missense variants were concentrated among those with hypertrophic cardiomyopathy and restrictive cardiomyopathy. Arrhythmogenic cardiomyopathy and dilated cardiomyopathy were characterized by a greater burden of ventricular arrhythmias, low QRS voltage, systolic dysfunction, and myocardial fibrosis, whereas hypertrophic cardiomyopathy and restrictive cardiomyopathy showed greater wall thickness and more atrial arrhythmias. Twelve pathogenic/likely pathogenic variants absent from gnomAD and ClinVar were identified. Ancestry analysis confirmed heterogeneous genetic backgrounds but detected no ancestry-related differences in variant class or phenotype. In exploratory outcome analyses, nonsustained ventricular tachycardia, low QRS voltage, and late gadolinium enhancement were associated with outcomes.

CONCLUSIONS: This deeply phenotyped Brazilian cohort expands the *FLNC* variant spectrum and supports broad genotype-phenotype associations across an admixed population. Exploratory findings suggest that markers of ventricular arrhythmia and myocardial fibrosis may complement left ventricular ejection fraction in future risk-stratification strategies, although validation in larger, independent cohorts is required.

Key Words: cardiomyopathy ■ genetic diversity ■ genotype-phenotype correlation ■ precision medicine ■ ventricular arrhythmia

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Nonstandard Abbreviations and Acronyms

ACM	arrhythmogenic cardiomyopathy
CMR	cardiac magnetic resonance
DCM	dilated cardiomyopathy
FLNC	filamin C
HCM	hypertrophic cardiomyopathy
HR	hazard ratio
ICD	Implantable Cardioverter Defibrillator
LVEF	left ventricular ejection fraction
RCM	restrictive cardiomyopathy
SCD	sudden cardiac death
VT	ventricular tachycardia

Filamin C (*FLNC*)-related cardiomyopathies are increasingly recognized as a distinct subset of inherited heart diseases. Variants in *FLNC*, which encodes a large actin-crosslinking and signaling scaffold protein localized at the sarcomeric Z-disk and intercalated disc, disrupt the mechanical integrity and mechano-signal transduction essential for cardiomyocyte homeostasis.^{1–3} Pathogenic or likely pathogenic (LP) *FLNC* variants cause a broad spectrum of cardiomyopathies, including dilated (DCM), arrhythmogenic (ACM), hypertrophic (HCM), and restrictive (RCM). Individuals with cardiomyopathy associated with truncating *FLNC* variants have a higher risk of adverse outcomes, including malignant ventricular arrhythmias and progressive heart failure.^{2–5} Two major molecular mechanisms underlie *FLNC* disease: haploinsufficiency resulting from truncating variants, which leads to sarcomere destabilization and cytoskeletal fragility, and toxic protein accumulation from missense variants.^{6–8} Truncating variants predominate in ACM and DCM, while missense variants are more common in HCM and RCM.^{9–11}

Large multicenter registries have estimated that 1% to 4% of patients with DCM carry pathogenic/LP *FLNC* variants, with many experiencing early ventricular arrhythmias or sudden cardiac death (SCD) despite only mild systolic impairment.^{5–9,12,13} *FLNC* truncating variants are now recognized as high-risk genetic markers, with potential implications for primary prevention strategies, including Implantable Cardioverter Defibrillator (ICD) placement, even when left ventricular ejection fraction (LVEF) is between 35% and 45%.^{1,2}

Most *FLNC* studies have focused on European and North American cohorts. Previous studies have demonstrated that the Brazilian population is characterized by high genetic admixture of European, African, and Native American ancestries, offering a unique opportunity to investigate the influence of ancestry on phenotypic expression.¹⁴

This study therefore aims to characterize genotype-phenotype associations, identify previously unreported variants, and describe clinical outcomes in a Brazilian *FLNC*-related cardiomyopathy cohort. Understanding

FLNC-related cardiomyopathies in Brazilian patients may uncover novel variants, refine clinical characterization, and improve the global representation of genetic variant data in precision medicine databases.

METHODS

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Study Design and Population

The data supporting the findings of this study are available from the corresponding author upon reasonable request. Detailed methods and statistical analysis code are provided in the [Supplemental Material](#). This study was approved by the Research Ethics Committee of the Heart Institute (InCor) of the University of São Paulo School of Medicine, São Paulo, Brazil (CAAE: 55200322.5.2001.0068; SDC 5440/22/019; approved on July 12, 2025). All participants provided informed consent in accordance with institutional requirements.

This cohort study was conducted within the MAPA Genoma Brasil initiative, a multicenter genomic registry designed to investigate genetic determinants of cardiomyopathies in the Brazilian population. Overall, 2654 unrelated probands with clinical diagnoses of genetic cardiomyopathies were consecutively enrolled for exome sequencing from tertiary cardiovascular centers including the Heart Institute (InCor) and collaborating hospitals across Brazil. Secondary cardiomyopathies due to Chagas disease, ischemic, inflammatory, or valvular disease were excluded in this first step.

Following identification of probands undergoing cardiomyopathy genetic testing who were carriers of *FLNC* pathogenic/LP variants, clinical and genetic familial cascade screening was performed in family members who agreed to participate. Demographic, clinical, imaging, and genetic data were analyzed. Patients were allocated to ACM, DCM, HCM, or RCM groups according to the main or first phenotype characterized.

Molecular Genetic Analysis

Genetic testing was performed by next-generation sequencing and Sanger sequencing. For this study, we included only variants classified as pathogenic/LP according to the American College of Medical Genetics and Genomics criteria. Two independent clinical geneticists interpreted variants according to American College of Medical Genetics and Genomics/Association for Molecular Pathology 2015 criteria.¹⁵ Probands and available family members who were carriers of pathogenic/LP *FLNC* variants were included in the registry. Peripheral blood DNA was extracted using the QIAamp DNA Blood Midi Kit (Qiagen, Germany). Exome sequencing was performed using next-generation sequencing with the Illumina NovaSeq 6000 platform (Twist kit), achieving mean on-target coverage >100× and >99% of bases covered at ≥20×. Reads were aligned to the GRCh38 reference genome, and variant calling was performed using DRAGEN (v3.10.9).

Genetic Ancestry Analysis

Genetic ancestry analysis was performed as a subgroup analysis and was restricted to individuals for whom blood samples and adequate DNA material were available in our laboratory

at the time of analysis. Whole-exome sequencing was performed in 44 subjects with clinically confirmed *FLNC*-related cardiomyopathies. A curated reference data set comprising 3007 genomes from the 1000 Genomes Project was used. Associations between variables were assessed using 3 ancestry frameworks (major ancestry, degree of admixture, and number of ancestry components), with statistical significance evaluated using Fisher exact, χ^2 , or Kruskal-Wallis tests as appropriate. Details of the genetic ancestry analysis are provided in the [Supplemental Methods](#).

Phenotypic Characterization

Demographic, clinical, imaging, and family history data were collected at study entry, from the MAPA Genoma Brasil electronic registry. Clinical evaluation included 12-lead ECG, transthoracic echocardiography, and, when available, 24-hour Holter and cardiac magnetic resonance (CMR) studies.^{16–18} Echocardiographic parameters, including diameters, wall thicknesses and LVEF, were obtained according to standardized imaging reports.¹⁹ Complete clinical, ECG, echocardiography, and longitudinal data were available for all phenotype-positive cohorts; CMR data were available for 49 patients and Holter monitoring for 34 patients.

Phenotypes were classified based on current international consensus guidance. The phenotype of DCM was defined as the presence of left ventricular dilatation and systolic dysfunction; RCM was defined as restrictive pathophysiology in the presence of normal ventricular volumes and normal ventricular wall thickness; HCM was defined as increased left ventricular wall thickness, all according to ESC cardiomyopathy guidelines.¹ The phenotype categories of ACM were defined according to the 2019 expert consensus,² considering an ACM phenotype when complex ventricular arrhythmias, such as a high burden of premature ventricular contractions, ventricular tachycardia (VT), nonsustained VT, and aborted sudden cardiac arrest, were the main clinical manifestations. The presence of only isolated ventricular ectopy did not characterize the patient as having an ACM phenotype. From a phenotypic standpoint, we observed a substantial degree of overlap and blended presentations within the ACM and DCM categories and also HCM and RCM. These cases were classified according to the most prominent phenotype after discussion among expert specialists. Patients were classified into 2 primary clinical categories based on phenotype: the ACM/DCM group if they presented with a phenotype of ACM, DCM, or a mixed phenotype of both ACM and DCM; or HCM/RCM group if they presented with a phenotype of HCM, RCM or a mixed phenotype of both HCM and RCM.

Study Outcomes

The study outcomes were: (1) arrhythmic outcomes: SCD or major ventricular arrhythmias. Major ventricular arrhythmias included sustained VT, ventricular fibrillation, aborted cardiac arrest, and appropriate ICD therapy; (2) heart failure outcomes: heart transplantation, or nonarrhythmic death, including heart failure death and death not due to SCD; and (3) composite end point: any arrhythmic or heart failure outcome. For outcome predictor analyses, patients with aborted cardiac arrest before follow-up initiation and phenotype-negative carrier relatives were excluded. The follow-up period for analysis ended at the date of the first outcome or at the last available contact

with the patient. VT was considered sustained when it lasted more than 30 seconds or was associated with hemodynamic instability. Appropriate ICD interventions included ICD shock or antitachycardia pacing for sustained VT. SCD was defined as sudden death with or without documented ventricular fibrillation or death within 1 hour of acute symptoms. The survival curves were based on time zero defined as the date of birth and included all carriers.

Statistical Analysis

Continuous variables were reported as mean $\pm$ SD or median with interquartile range, according to data distribution. Categorical variables were presented as frequencies and percentages. Between-group comparisons used the χ^2 or Fisher exact test for categorical variables and the Student *t* test or Mann-Whitney *U* test for continuous variables, as appropriate. Age-based event-free survival was evaluated using Kaplan-Meier curves, with birth defined as time zero and censoring at the last available clinical evaluation. Exploratory Cox proportional hazards analyses were performed separately for arrhythmic, heart failure, and composite outcomes. Because of the limited number of events and incomplete availability of some imaging and rhythm-monitoring variables, each covariate was evaluated in a separate univariable model. Hazard ratios (HRs), 95% CIs, nominal *P* values, and false discovery rate-adjusted *P* values were reported. A 2-sided *P* < 0.05 was considered statistically significant.

Data Availability

Deidentified participant data and analytic scripts supporting the findings are available through the MAPA Genoma Brasil data repository upon reasonable request and approval of a data use agreement, in accordance with institutional policies and Brazilian data protection regulations.

RESULTS

Cohort Overview

The study included 97 carriers of pathogenic/LP *FLNC* variants, with a median age of 33 years (16–43 years); 49% were male and 58 (60%) were probands. Among 2654 unrelated probands enrolled in the registry, 58 carried a pathogenic/LP *FLNC* variant, corresponding to a diagnostic yield of 2.2%. ACM was the most frequent phenotype, affecting 46 carriers, followed by dilated cardiomyopathy in 17, restrictive cardiomyopathy in 6, and hypertrophic cardiomyopathy in 3. Among the 39 genotype-positive relatives, aged 5 to 67 years, 25 did not fulfill specific cardiomyopathy phenotype criteria at the time of evaluation. Of the 14 phenotype-positive relatives, 12 had ACM, 1 had DCM, and 1 had RCM.

Among probands, the most common presenting symptoms were dyspnea (67%) and palpitations (55%), whereas syncope (14%) and aborted cardiac arrest (10%) were less frequent. Most probands were in sinus rhythm. Atrial fibrillation was present in 10%; 1 patient had advanced atrioventricular block, and 39% had left

Table 1. Demographic and Baseline Clinical Characteristics of *FLNC* Pathogenic/LP Variant Carriers

	Probands (n=58)		Relatives with the variant (n=39)		All carriers (n=97)	
	Evaluated	Finding	Evaluated	Finding	Evaluated	Finding
Age at diagnosis	58	35 (26–43)	39	27 (11–40)	97	33 (16–43)
Variant type, n (%)						
Missense	58	8 (14)	39	1 (3)	97	9 (9)
Truncating	58	50 (86)	39	38 (97)	97	88 (91)
Phenotype, n (%)						
ACM	58	34 (59)	39	12 (31)	97	46 (47)
DCM	58	16 (28)	39	1 (3)	97	17 (17)
HCM	58	3 (5)	39	0 (0)	97	3 (3)
RCM	58	5 (9)	39	1 (3)	97	6 (6)
No phenotype	58	0 (0)	39	25 (64)	97	25 (26)
Male, n (%)	58	29 (50)	39	19 (49)	97	48 (49)
Symptoms, n (%)						
Palpitations	58	32 (55)	39	7 (18)	97	39 (40)
Dyspnea	58	39 (67)	39	10 (26)	97	49 (50)
Chest pain	58	14 (24)	39	6 (15)	97	20 (20)
Syncope	58	8 (14)	39	0 (0)	97	8 (8)
Aborted cardiac arrest	58	6 (10)	39	0 (0)	97	6 (6)
ECG, n (%)						
Advanced atrioventricular block	57	1 (2)	38	0 (0)	95	1 (1)
Right bundle branch block	57	1 (2)	38	4 (10)	95	5 (5)
Left bundle branch conduction disorder	57	22 (39)	38	6 (16)	95	28 (30)
Low voltages	57	24 (42)	38	15 (39)	95	39 (41)
Left ventricular overload	57	3 (5)	38	2 (5)	95	5 (5)
Negative T-wave	56	35 (63)	38	10 (26)	95	45 (47)
Negative T-wave in V1–V3	57	11 (19)	38	4 (10)	95	15 (16)
Negative T-wave in V4–V6	57	21 (37)	38	4 (10)	95	25 (26)
Negative T-wave in DI-AVL	57	21 (37)	38	3 (8)	95	24 (25)
Negative T-wave DII, DIII, AVF	57	10 (18)	38	2 (5)	95	12 (13)
Arrhythmias, n (%)						
EEVV	58	47 (81)	37	14 (38)	95	61 (64)
TVNS	39	24 (61)	11	1 (10)	50	25 (50)
Atrial fibrillation	57	5 (9)	38	1 (2)	95	6 (6)
Pacemaker	57	2 (3)	38	1 (2)	95	3 (3)
Echocardiogram						
IV septum (mm)	56	9 (7–10)	36	8 (6–8)	92	8 (7–9)
LVEF (%)	58	31 (25–53)	36	60 (49–60)	94	46 (27–60)
RV dysfunction, n (%)	57	14 (25)	36	2 (6)	93	16 (17)
Wall-motion abnormalities, n (%)	57	16 (28)	35	6 (17)	92	22 (24)
Cardiac magnetic resonance						
RVEF (%)	38	46.7 (±2.3)	9	55.3 (±2.7)	47	48.3 (±2.0)
RVEDV (mL)	36	98 (69–117)	7	101.3 (±12.4)	43	98 (71–115)
RVESV (mL)	36	47 (29–79)	7	57.0 (±12.2)	43	47 (29–80)
LVEF (%)	41	37 (24–53)	8	41 (34–49)	49	38.8 (±2.2)
LVEDV (mL)	40	142 (93–173)	8	135 (101–151)	48	142 (95–173)
LVESV (mL)	40	93 (53–127)	8	94 (53–100)	48	93 (53–117)
Left atrium (mm)	36	40 (31–51)	8	31.5 (±2.7)	44	40 (30–46)

(Continued)

Table 1. Continued

	Probands (n=58)		Relatives with the variant (n=39)		All carriers (n=97)	
	Evaluated	Finding	Evaluated	Finding	Evaluated	Finding
IV septum (mm)	39	8 (7–9)	8	8 (6–8)	47	8 (7–9)
Posterior wall thickness (mm)	39	8 (7–9)	8	8 (6–8)	47	8 (7–9)
Maximum thickness (mm)	40	8 (7–10)	8	8 (6–8)	48	8 (7–9)
RV wall-motion abnormality, n (%)	41	14 (34)	8	0 (0)	49	14 (29)
LGE, n (%)	42	31 (74)	7	4 (57)	49	35 (71)
Ring-like pattern, n (%)	27	14 (52)	5	3 (60)	32	17 (53)
Outcomes, n (%)						
VF/VT	58	13 (22)	39	0 (0)	97	13 (13)
ICD shock	58	4 (7)	39	0 (0)	97	4 (4)
SCD	58	1 (2)	39	1 (3)	97	2 (2)
Heart failure hospitalization	58	20 (34)	39	1 (3)	97	21 (22)
Heart transplantation	58	8 (14)	39	1 (3)	97	9 (9)
Nonarrhythmic death	58	8 (14)	39	0 (0)	97	8 (8)

Continuous variables are presented as median (interquartile range); categorical variables are presented as n (%). ACM indicates arrhythmogenic cardiomyopathy; AF, atrial fibrillation; CMR, cardiac magnetic resonance; DCM, dilated cardiomyopathy; ECG, electrocardiography; HCM, hypertrophic cardiomyopathy; ICD, Implantable Cardioverter Defibrillator; LGE, late gadolinium enhancement; LP, likely pathogenic; LVEF, left ventricular ejection fraction; NSVT, nonsustained ventricular tachycardia; PVC, premature ventricular complexes; RCM, restrictive cardiomyopathy; RVEF, right ventricular ejection fraction; SCD, sudden cardiac death; VF, ventricular fibrillation; and VT, ventricular tachycardia.

bundle branch conduction abnormalities. T-wave inversion was observed in 63% of probands, most commonly in the lateral precordial leads V4 to V6 (37%). No patient had isolated T-wave inversion in the right precordial leads, and no epsilon waves were identified. Low QRS voltage in the limb leads was present in 41%.

Ventricular arrhythmias were documented in 81% of probands, including frequent premature ventricular complexes exceeding 500 beats per 24 hours. Among patients with available Holter monitoring, nonsustained VT was detected in 61%. Left ventricular systolic dysfunction, defined as LVEF <55%, was present in 43 of 58 probands (74%), and left ventricular dilation was observed in 45%. Right ventricular structural or wall-motion abnormalities, including dilation, akinesia, or dyskinesia, were present in 34%. Among the 42 probands who underwent CMR, 31 (74%) had late gadolinium enhancement, and 14 showed a ring-like enhancement pattern. Baseline characteristics of the overall cohort, stratified by proband status, are presented in Table 1.

Genetic and Phenotypic Characteristics

Among the 97 carriers of pathogenic/LP *FLNC* variants, 88 carried truncating variants and 9 carried missense variants. Among the 58 probands, 29 distinct pathogenic/LP *FLNC* variants were identified, including 24 truncating and 5 missense variants. The truncating variants comprised 7 canonical splice-site variants, 8 frameshift variants, 8 nonsense variants, and 1 multiexon deletion, consistent with a haploinsufficiency mechanism. One

patient also carried an additional pathogenic/LP variant in another cardiomyopathy-associated gene, *DSC2* c.1053_1059del (p.His351Glnfs*9), and presented with biventricular ACM. The distribution and classification of the identified variants are presented in Table 2 and Figure 1.

Phenotype-specific characteristics are summarized in Table 3. Among patients with ACM, T-wave inversion and low QRS voltage were present in 67% and 60%, respectively. Left ventricular systolic function was impaired in both the ACM and DCM groups, with median LVEF values of 38% and 29%, respectively. Late gadolinium enhancement was detected in 29 of 34 patients with ACM who underwent CMR (85%).

Comparisons between the combined phenotype groups are presented in Table 4. Truncating variants were substantially more frequent among patients with ACM/DCM phenotypes than among those with HCM/RCM phenotypes (62 of 63 [98%] versus 1 of 9 [11%]), whereas missense variants predominated in the HCM/RCM group (8 of 9 [89%]; *P*<0.001). Compared with patients with HCM/RCM, those with ACM/DCM had a higher prevalence of ventricular arrhythmias (50 of 63 versus 0 of 9; *P*<0.001) and low QRS voltage (32 of 63 versus 0 of 9; *P*=0.002), lower LVEF (31% versus 62%; *P*=0.001), and lower right ventricular ejection fraction (48% versus 60%; *P*=0.009). In contrast, patients with HCM/RCM had a higher prevalence of atrial arrhythmias (4 of 9 versus 4 of 63; *P*=0.011), greater maximum left ventricular wall thickness (16 versus 8 mm; *P*<0.001), and more frequent diastolic dysfunction (*P*=0.025).

Table 2. Pathogenic and Likely Pathogenic FLNC Variants Identified in the Cohort

Exon	Nucleotide change	Protein consequence	Variant class	ACMG classification	ACMG criteria	No. of probands	New variant
33	c.5520T>A	p.Tyr1840*	Nonsense	Pathogenic	PVS1, PS4, PM2_supporting	13	No
41	c.6976C>T	p.Arg2326*	Nonsense	Pathogenic	PVS1, PS4_moderate, PM2, PP1	9	No
4 (splice donor)	c.850+1G>A	p.(?)	Splice-site	Likely pathogenic	PVS1, PM2	2	No
41	c.6889G>A	p.Val2297Met	Missense	Pathogenic	PS3_supporting, PS4_moderate, PM2_supporting, PP1, PP4	2	No
34	c.5613_5635del	p.Gly1872Leufs*28	Frameshift deletion	Pathogenic	PVS1, PM2, PS4_supporting	2	No
27	c.4636G>A	p.Gly1546Ser	Missense	Likely pathogenic	PS4_supporting, PM2_supporting, PP1_strong, PP3	2	No
27 (splice acceptor)	c.4581-1G>A	p.(?)	Splice-site	Likely pathogenic	PVS1, PM2	2	Yes
23	c.4079T>G	p.Leu1360Arg	Missense	Likely pathogenic	PS4_moderate, PM2, PP3	2	Yes
22 (splice donor)	c.3964+1G>T	p.(?)	Splice-site	Likely pathogenic	PVS1, PM2	2	No
NA	Deletion of exons 2-8	X	Multiexon deletion	Pathogenic	2E (0.45)+4F (0.15)+5D (0.15)	1	No
48	c.8053A>T	p.Lys2685*	Nonsense	Likely pathogenic	PVS1, PM2	1	Yes
46 (splice acceptor)	c.7562-1G>A	p.(?)	Splice-site	Likely pathogenic	PVS1, PM2	1	No
39	c.6462T>A	p.Cys2154*	Nonsense	Likely pathogenic	PVS1, PM2	1	Yes
18	c.5814_5823del	p.Gly1939Serfs*11	Frameshift deletion	Likely pathogenic	PVS1, PM2	1	Yes
35	c.5558_5559del	p.Tyr1853Cysfs*54	Frameshift deletion	Pathogenic	PVS1, PM2, PP1	1	Yes
33	c.5405_5408dup	p.Met1804Alafs*28	Frameshift duplication	Pathogenic	PVS1, PM2, PP1	1	Yes
2	c.487C>T	p.Gln163*	Nonsense	Likely pathogenic	PVS1, PM2	1	Yes
28	c.4871C>T	p.Ser1624Leu	Missense	Likely pathogenic	PS3_supporting, PS4_moderate, PM2_supporting, PP3, PP1	1	No
28	c.4846dup	p.Tyr1616Leufs*4	Frameshift duplication	Likely pathogenic	PVS1, PM2	1	Yes
26 (splice donor)	c.4580+1G>A	p.(?)	Splice-site	Pathogenic	PVS1, PM2, PP1	1	No
21	c.3557C>T	p.Ala1186Val	Missense	Likely pathogenic	PS4, PM2, PM6	1	No
20	c.3014C>A	p.Ser1005*	Nonsense	Likely pathogenic	PVS1, PM2	1	No
19	c.2813del	p.Gly938Alafs*5	Frameshift deletion	Likely pathogenic	PVS1, PM2	1	No
14	c.2263del	p.Arg755Glyfs*2	Frameshift deletion	Likely pathogenic	PVS1, PM2	1	Yes
12	c.1849G>T	p.Glu617*	Nonsense	Likely pathogenic	PVS1, PM2	1	No
11 (splice donor)	c.1813+1G>A	p.(?)	Splice-site	Likely pathogenic	PVS1, PM2	1	Yes
10 (splice donor)	c.1676+1G>A	p.(?)	Splice-site	Likely pathogenic	PVS1, PM2	1	No
10	c.1579G>T	p.Glu527*	Nonsense	Likely pathogenic	PVS1, PM2	1	Yes
8	c.1260del	p.Arg421Glyfs*71	Frameshift deletion	Likely pathogenic	PVS1, PM2	1	No

Data presented include nucleotide change, protein consequence, variant class (nonsense, splice site, frameshift deletion/duplication, missense, deletion), ACMG classification (pathogenic or likely pathogenic), ACMG criteria applied, GnomAD allele frequency, maximum populational frequency in GnomAD, REVEL score, SIFT prediction, ClinVar significance, and whether the variant is novel. ACMG indicates American College of Medical Genetics.

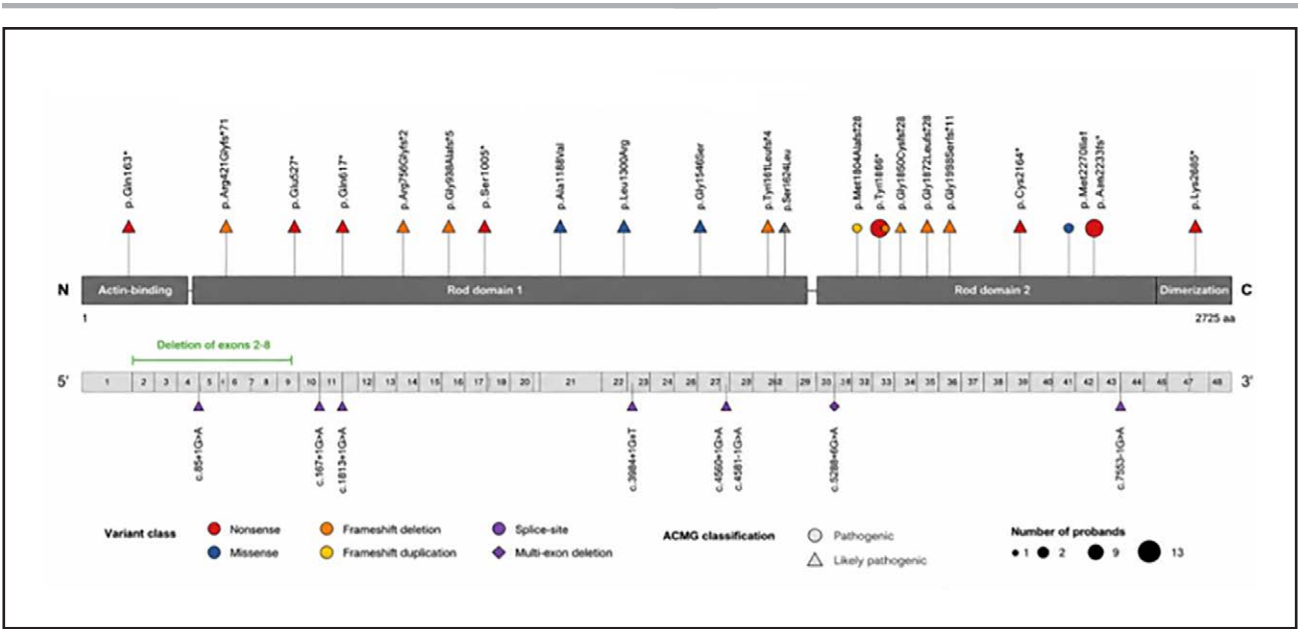


Figure 1. Genetic variant distribution across the filamin C (*FLNC*) gene. Lollipop plot of the *FLNC* gene structure showing the distribution of 29 distinct pathogenic/likely pathogenic (LP) variants identified in 58 probands. Variants are mapped to the corresponding protein domains, including the actin-binding domain, rod domains 1 and 2, and the dimerization domain. Each lollipop represents a unique variant and is color-coded according to variant class: nonsense (red), frameshift deletion (orange), frameshift duplication (yellow), missense (blue), splice-site (purple), and multiexon deletion (violet). The shape at the tip of each lollipop indicates American College of Medical Genetics and Genomics (ACMG) classification: circle, pathogenic; triangle, LP. The lower illustrates the genomic location of splice-site variants and the deletion of exons 2–8. VUS indicates variant of uncertain significance.

Genetic Ancestry Analysis

Genetic ancestry was evaluated in 44 *FLNC* variant carriers with available exome data. Principal component analysis showed a broad distribution along a continuum between European, African, and Amerindian reference populations, consistent with the admixed structure of the Brazilian population, although most individuals clustered closer to the European reference group (Figure S1). One participant, CMG1098, had a predominantly African ancestry component, defined as >50% contribution from a single reference population, whereas CMG6054 had a predominantly Amerindian ancestry component.

Individual ancestry-proportion estimates were consistent with the principal component analysis and demonstrated substantial interindividual diversity (Figure S2). Twenty participants had contributions of at least 10% from 3 or more ancestry components, and 17 had contributions of at least 10% from 2 ancestry components. The remaining participants had 1 predominant ancestry component according to this threshold.

After correction for multiple comparisons, no statistically significant associations were identified between the ancestry measures and phenotype, cDNA variant, protein consequence, variant class, or truncating versus missense status. False discovery rate–adjusted *P* values ranged from 0.75 to 1.00 (Table S1).

Ancestry analyses were performed only in a subset of the overall cohort (44 individuals); given the limited size

of the ancestry subset, although these findings provide preliminary insights into the relationship between ancestry and *FLNC*-related cardiomyopathies, these findings should be considered exploratory.

Cohort Outcomes

All arrhythmic outcomes occurred in patients with an ACM phenotype. During follow-up, 13 patients experienced VT, 4 received appropriate ICD shocks, and 2 died suddenly. Because some patients experienced more than 1 arrhythmic event, these categories were not mutually exclusive.

The 2 patients who experienced SCD were male carriers of truncating *FLNC* variants. One was a 30-year-old proband carrying c.6976C>T (p.Arg2326*), with a family history of SCD and an LVEF of 48%. The other was a 41-year-old genotype-positive relative carrying the canonical splice-site variant c.1676+1G>A, with an LVEF of 36%.

Aborted cardiac arrest was the initial clinical manifestation in 6 probands. Their mean age at presentation was 33 years, 67% were male, and mean LVEF was 39% (range, 26% to 59%). Notably, 6 patients who experienced SCD or a major ventricular arrhythmia had preserved or only mildly reduced systolic function, with LVEF values ranging from 48% to 67%.

Seventeen patients reached a heart failure end point. Nine underwent heart transplantation, and 8 died from nonarrhythmic causes. An overview of cohort

Table 3. Clinical Characteristics According to Phenotypes

Variable	ACM (n=46)	DCM (n=17)	HCM (n=3)	RCM (n=6)	P value
Proband n (%)	34 (74)	16 (94)	3 (100)	5 (83)	0.163
Age at symptoms, y (mean $\pm$ SD)	38 (30–44)	35 (21–50)	34 (2–40)	24 (16–33)	0.196
Male sex, n (%)	28 (61)	7 (41)	2 (67)	2 (33)	0.356
Genetic variant type, n (%)					
Truncating	46 (100)	16 (94)	1 (33)	0 (0)	<0.001*
Missense	0 (0)	1 (6)	2 (67)	6 (100)	<0.001*
Family history of severe heart failure or heart transplant or sudden death, n (%)	19 (41)	6 (35)	1 (33)	3 (50)	0.921
Symptoms at diagnosis, n (%)					
Palpitations	26 (56)	9 (53)	0 (0)	2 (33)	0.131
Syncope	4 (9)	2 (12)	2 (67)	0 (0)	0.07
Recovered cardiac arrest	5 (11)	1 (6)	0 (0)	0 (0)	0.558
Arrhythmias					
Atrial arrhythmia	3 (6)	1 (6)	1 (33)	3 (50)	0.04*
Ventricular arrhythmia	46 (100)	4 (23)	0 (0)	0 (0)	<0.001*
ECG, n (%)					
Conduction system disease	6 (13)	5 (29)	1 (33)	0 (0)	0.191
T-wave inversion	30 (67)	5 (29)	1 (50)	2 (33)	0.041*
Low QRS voltage	27 (60)	5 (29)	0 (0)	0 (0)	0.001*
Cardiac imaging					
ECHO – LVEF (%), median (IQR)	38 (25–52)	29 (23–42)	63 (50–67)	61 (55–65)	<0.001*
CMR with LGE, n (%)	29 (85)	3 (37)	2 (100)	1 (25)	0.006*
CMR – LVEF (%), median (IQR)	35 (25–48)	22 (16–47)	56 (55–57)	60 (60–69)	0.006*
CMR – RVEF (%), median (IQR)	50 (42–58)	39 (21–57)	60 (58–62)	60 (51–60)	0.139
LV maximum thickness (mm)	8 (7–9)	8 (7–10)	18 (17)	12 (11–12)	0.006*

Continuous variables are presented as median (IQR); categorical variables are presented as n (%). Statistical significance was determined using Kruskal-Wallis test for continuous variables and χ^2 or Fisher Exact test for categorical variables. $P < 0.05$ were considered statistically significant. ACM indicates arrhythmogenic cardiomyopathy; CMR, cardiac magnetic resonance; DCM, dilated cardiomyopathy; ECG, electrocardiography; ECHO, echocardiography; HCM, hypertrophic cardiomyopathy; IQR, interquartile range; LGE, late gadolinium enhancement; LV, left ventricular; LVEF, left ventricular ejection fraction; RCM, restrictive cardiomyopathy; and RVEF, right ventricular ejection fraction.

ascertainment, phenotype distribution, and major clinical outcomes is presented in Figure 2.

Exploratory Outcome Analyses

For exploratory outcome analyses, 31 of the 97 FLNC variant carriers were excluded: 25 phenotype-negative relatives and 6 patients whose aborted cardiac arrest occurred before the beginning of follow-up. The resulting outcome cohort included 66 individuals; analyses were also performed separately in the proband subgroup (n=52). Because of the limited sample size and event count, univariable Cox proportional hazards models were used to evaluate associations between clinical, electrocardiographic, imaging, and genetic variables and arrhythmic, heart failure, and composite outcomes (Figure 3). The proportional hazards assumption was not violated in any of the 54 covariate-outcome combinations (all $P > 0.05$, ranging from 0.12 to 0.98).

Nonsustained VT was associated with arrhythmic outcomes in both the overall outcome cohort (HR, 2.76 [95% CI, 1.51–5.03]; false discovery rate-adjusted $P < 0.01$) and the proband subgroup (HR, 3.01 [95% CI, 1.75–5.17]; false discovery rate-adjusted $P < 0.001$). The presence of late gadolinium enhancement on CMR imaging was associated with the composite outcome in the overall cohort (HR, 2.55 [95% CI, 1.42–4.57]) and in the proband subgroup (HR, 2.09 [95% CI, 1.28–3.42]). Low QRS voltage was also associated with adverse outcomes across the evaluated models. In the proband subgroup, low QRS voltage (HR, 2.86 [95% CI, 1.39–5.89]), T-wave inversion (HR, 2.70 [95% CI, 1.41–5.16]), and premature ventricular complexes (HR, 2.66 [95% CI, 1.42–4.97]) were associated with arrhythmic outcomes.

Truncating variants were associated with a higher rate of arrhythmic outcomes than missense variants in the proband subgroup (HR, 2.28 [95% CI, 1.14–4.58]; false discovery rate-adjusted $P = 0.05$). These associations should be interpreted as exploratory because of

Table 4. Clinical and Imaging Characteristics According to Phenotypic Cluster

Variable	ACM/DCM (N=63)	HCM/RCM (N=9)	P value
Genetic variant type, n (%)			
Truncating	62 (98)	1 (11)	<0.001
Missense	1(2)	8 (89)	<0.001
Arrhythmias, n (%)			
Atrial arrhythmia	4 (6)	4 (40)	0.011
Ventricular arrhythmia	50 (79)	0	<0.001
ECG, n (%)			
Low QRS voltage	32 (52)	0	0.002
Cardiac imaging			
ECHO – LVEF (%), median (IQR)	31 (25–48)	62 (55–65)	0.001
CMR – LVEF (%), median (IQR)	34 (24–47)	60 (56–64)	<0.001
CMR – RVEF (%), median (IQR)	48 (34–58)	60 (56–63)	0.009
LV maximum thickness (mm)	8 (7–9)	16 (12–20)	<0.001

Continuous variables are presented as median (IQR); categorical variables are presented as n (%). Statistical significance was determined using Fisher Exact test (2x2 tables) or χ^2 test for larger tables. $P<0.05$ were considered statistically significant. ACM indicates arrhythmogenic cardiomyopathy; CMR, cardiac magnetic resonance; DCM, dilated cardiomyopathy; ECG, electrocardiography; ECHO, echocardiography; HCM, hypertrophic cardiomyopathy; IQR, interquartile range; ; LVEF, left ventricular ejection fraction; RCM, restrictive cardiomyopathy; and RVEF, right ventricular ejection fraction.

the limited number of events and variable availability of Holter and CMR data.

Survival Analysis

The estimated mean age free from the composite outcome did not differ significantly between carriers of truncating and missense *FLNC* variants (56.6 years [95% CI, 52.1–61.2] versus 52.2 years [95% CI, 36.5–67.9], respectively; log-rank $P=0.541$). Age-based Kaplan-Meier curves for composite, arrhythmic, and heart failure outcomes, stratified by variant class, are presented in Figure 4.

DISCUSSION

In the present study, we reported the comprehensive characterization of variants, phenotypes, and outcomes of *FLNC* cardiomyopathy in the first Latin American population cohort of *FLNC* carriers.

The identification of 12 novel variants broadens the *FLNC* variant landscape and emphasizes the value of incorporating genetically diverse populations in genomic medicine. The identification of novel variants in an admixed population has implications for variant reclassification efforts.

Our data align with findings from previous cohorts,^{3–5,20–22} which demonstrated that truncating *FLNC* variants represent a major monogenic cause of ACM/DCM with high SCD risk.^{20–22} Consistent with previous reports,^{3–5} we observed a predominance of truncating

variants, especially among patients with ACM/DCM phenotypes. Truncating variants were predominantly associated with increased ventricular arrhythmias, low QRS voltage, reduced LVEF, and right ventricular dysfunction. Nearly one-third of patients with ACM/DCM experienced arrhythmic outcomes, highlighting the malignant arrhythmic potential of *FLNC*-related disease. SCD/major ventricular arrhythmias occurred even in the absence of severe systolic dysfunction, including events in patients with LVEF >35% and in some with LVEF >50%. This observation is consistent with previous reports indicating that LVEF alone may not fully capture arrhythmic risk in *FLNC*-related cardiomyopathy.^{1,2,10}

Our findings reinforce the association between truncating *FLNC* variants and high-risk DCM and ACM, while also revealing the broader phenotypic variability of this gene, including hypertrophic and restrictive forms. These morphofunctional contrasts align with prior mechanistic models of *FLNC* disease.^{3,9,11–13}

Truncating *FLNC* variants were mainly associated with ACM/DCM, supporting haploinsufficiency as the primary mechanism. Loss of filamin C destabilizes the Z-disk and cytoskeleton, leading to cardiomyocyte injury and replacement fibrosis, clinically manifesting as left ventricular dilation, systolic dysfunction, late gadolinium enhancement, and ventricular arrhythmias.^{3–5,11} The ring-like late gadolinium enhancement pattern on CMR further suggests a fibrosis-driven arrhythmogenic substrate, similar to the desmosomal cardiomyopathies.^{3–5} In contrast, missense variants were more frequently observed in patients with HCM/RCM phenotypes, and were associated with a higher prevalence of atrial arrhythmias, diastolic dysfunction and myocardial wall thickness. These findings align with the proposed mechanistic dichotomy and prior reports.^{6–8,11–13}

Importantly, phenotype–genotype relationships are not absolute: 1 individual in our cohort carried a truncating variant yet presented with an HCM/RCM phenotype, whereas 1 individual carrying a missense variant exhibited an ACM/DCM phenotype. These exceptions suggest that additional genetic, epigenetic, or environmental modifiers may influence penetrance and phenotypic expression, highlighting the complexity of *FLNC* physiology beyond the haploinsufficiency and gain-of-function dichotomy.²⁰

Our cohort also demonstrated the phenotypic diversity of *FLNC*-related cardiomyopathies. We identified patients with blended phenotypes that did not fit neatly into a single traditional diagnostic category.¹ Such variability suggests that genetic modifiers, environmental influences, and coexisting variants in other cardiomyopathy-associated genes may modulate clinical expression.²⁰

Notably, several patients who experienced arrhythmic events had preserved or only mildly reduced LVEF, reinforcing the limitations of relying exclusively on

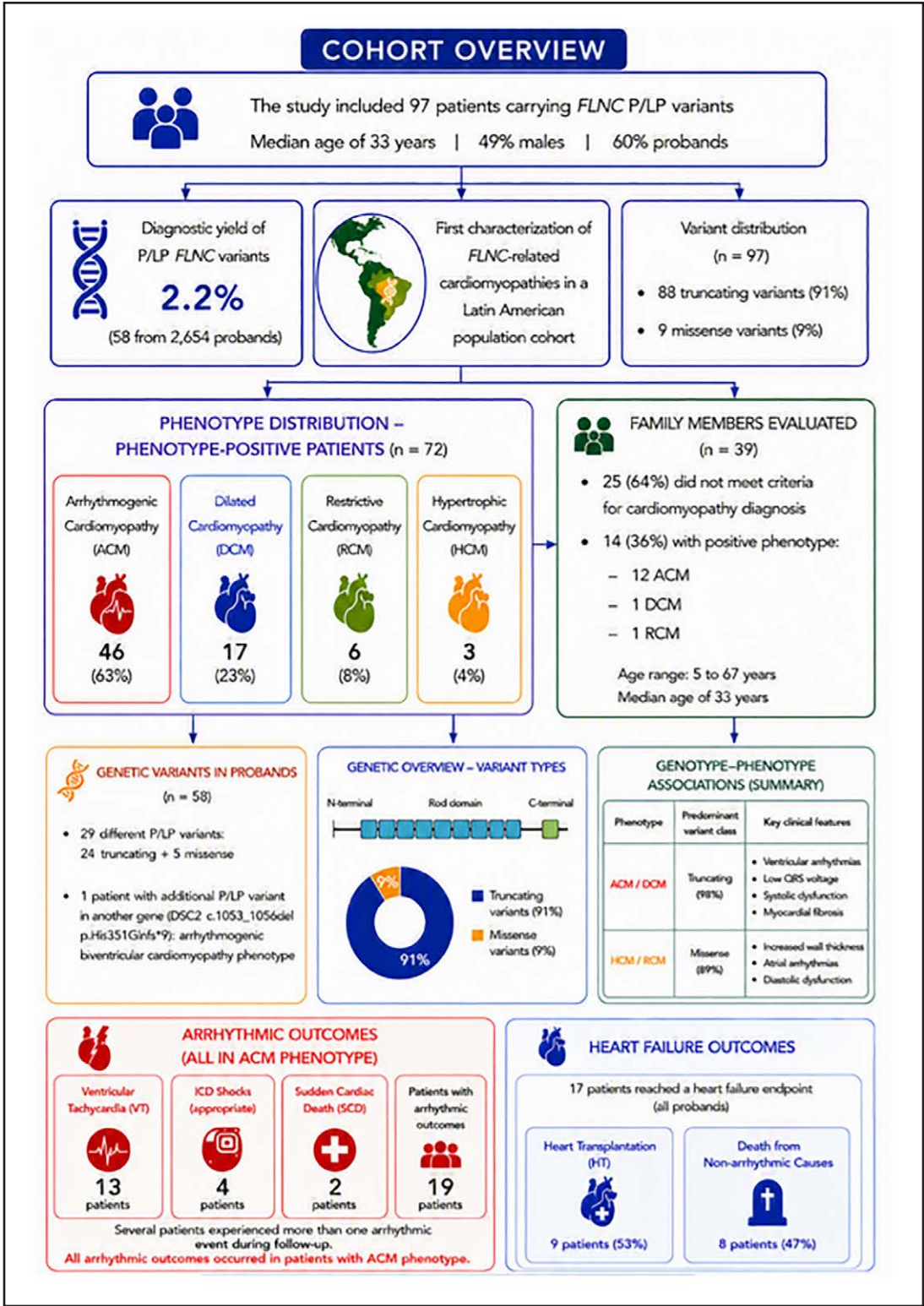


Figure 2. Flowchart summarizing the cohort composition, phenotype distribution, and major clinical outcomes of *FLNC* pathogenic/likely pathogenic (LP) variant carriers.

The study included 97 carriers of pathogenic/LP *FLNC* variants, comprising 58 probands and 39 family members identified through cascade screening. Arrhythmic cardiomyopathy (ACM) was the predominant phenotype, followed by dilated cardiomyopathy (DCM), restrictive cardiomyopathy (RCM), and hypertrophic cardiomyopathy (HCM). All arrhythmic outcomes occurred in patients with an ACM phenotype and included ventricular tachycardia (VT), appropriate Implantable Cardioverter Defibrillator (ICD) shocks, and sudden cardiac death (SCD). Heart failure outcomes included heart transplantation (HT) and nonarrhythmic death.

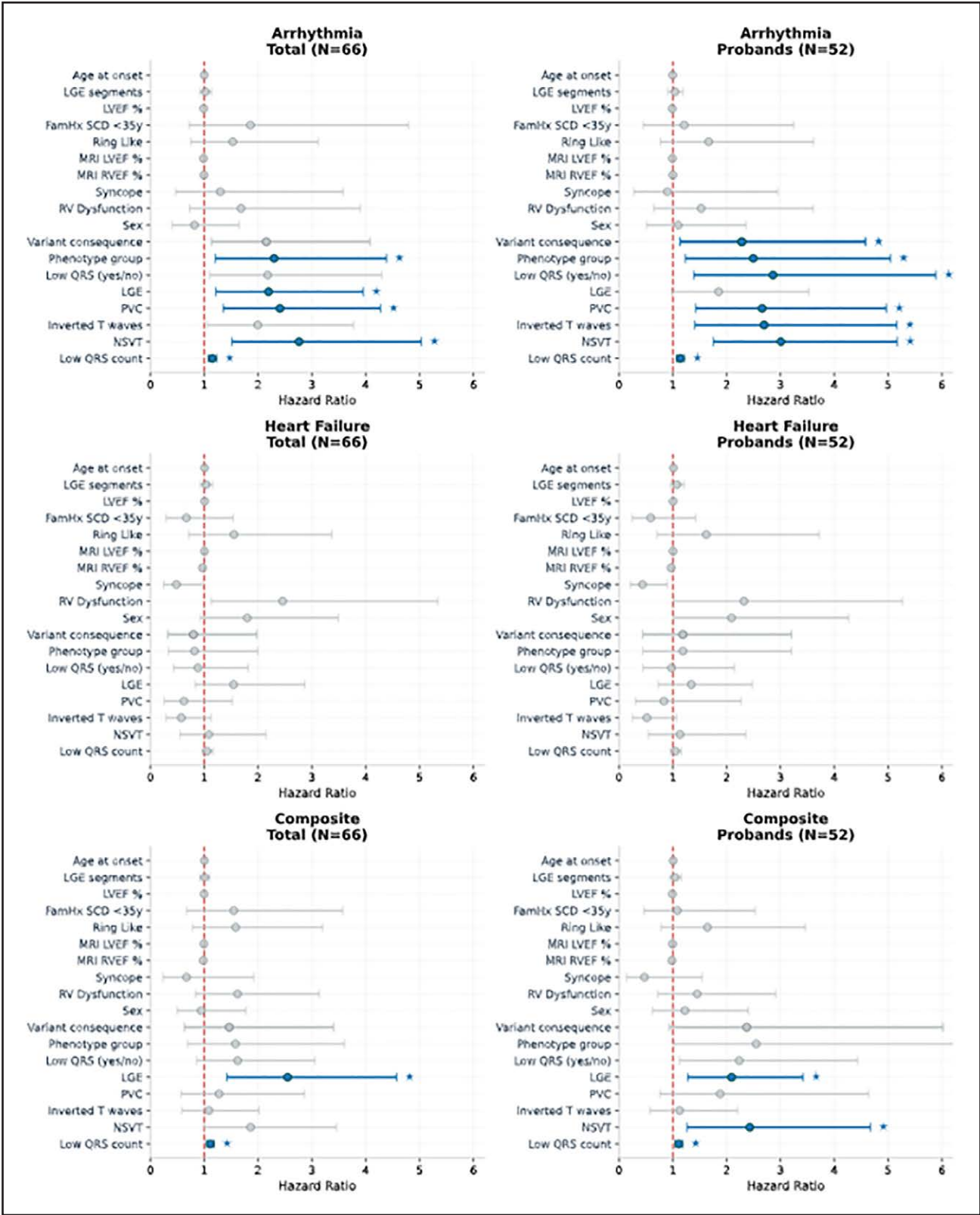


Figure 3. Forest plots of hazard ratios (HRs) for arrhythmic and composite clinical outcomes. Univariate Cox proportional hazards regression models for the total cohort (N=66, **left**) and the proband subcohort (N=52, **right**). **Top:** risk factors associated with arrhythmic events; **middle:** risk factors associated with heart failure; **bottom:** Risk factors associated with the composite clinical end point. Data points represent computed HRs, and horizontal error bars indicate corresponding 95% CIs. The vertical red dashed line marks the null effect (HR, 1.0). Blue lines and markers indicate variables that retain statistical significance after false discovery rate correction ($P<0.05$), marked with an asterisk (*). Gray lines and markers indicate variables that were not significant after multiple P value corrections.

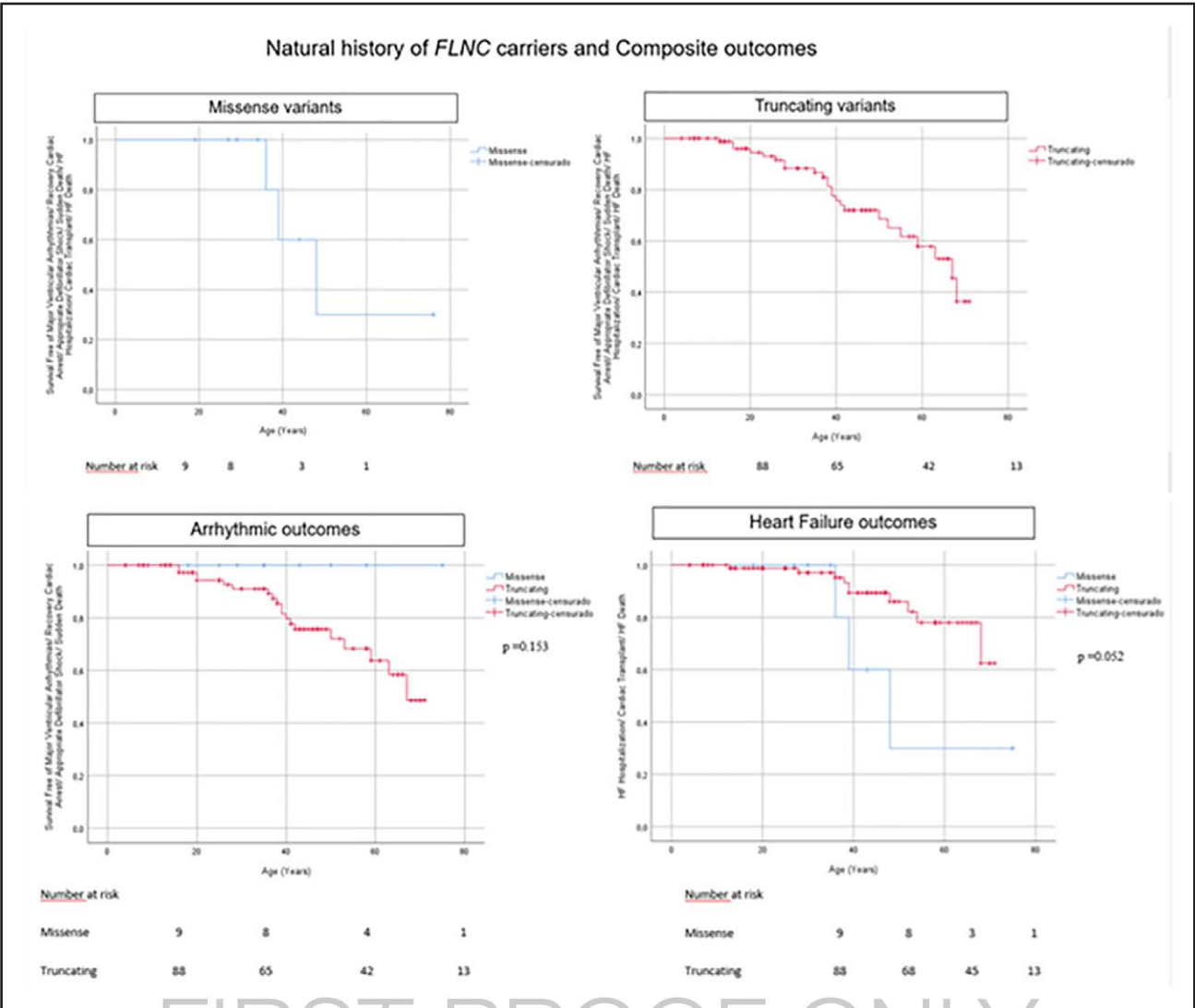


Figure 4. Age-based Kaplan-Meier curves for composite, arrhythmic, and heart failure outcomes according to *FLNC* variant class. Kaplan-Meier curves comparing carriers of missense and truncating *FLNC* variants. The **upper** show event-free survival for the composite outcome, and the **lower** show arrhythmic and heart failure event-free survival stratified by variant class. Tick marks indicate censored observations.

systolic function for risk stratification in *FLNC*-related cardiomyopathy.

Our exploratory outcome analyses provide additional context for the clinical characterization of this cohort. Nonsustained VT and low QRS voltage were associated with arrhythmic events, whereas the presence of late gadolinium enhancement was associated with the composite clinical outcome. These observations are biologically plausible and consistent with the concept that arrhythmic burden and replacement fibrosis contribute to the arrhythmogenic substrate of *FLNC*-related cardiomyopathy. LVEF was not significantly associated with outcomes in the present analyses. Future risk models should evaluate whether electrocardiographic, rhythm-monitoring, imaging, and genetic variables provide complementary prognostic information beyond conventional measures of ventricular function.

Several limitations should be considered. The cohort was assembled predominantly through tertiary cardiovascular centers and may therefore be enriched for more severe or clinically recognizable phenotypes. The relatively small number of patients, particularly those carrying missense variants or presenting with HCM/RCM phenotypes, limited statistical power and the precision of subgroup comparisons. Accordingly, these analyses should be regarded as exploratory and hypothesis-generating rather than as the derivation of a clinical prediction model. CMR and Holter monitoring were not available for all participants, and incomplete family participation prevented reliable estimation of age-dependent penetrance.

Genetic ancestry was available for only a subset of the cohort. Although this analysis documented heterogeneous ancestral contributions and did not identify significant associations, its limited sample size precludes

exclusion of ancestry-specific effects. The principal contribution of this analysis is therefore to demonstrate the inclusion of a genetically diverse population that remains underrepresented in cardiovascular genomic studies.

CONCLUSIONS

This deeply phenotyped Brazilian cohort broadens the known spectrum of pathogenic/LP *FLNC* variants and supports the general association of truncating variants with ACM and DCM phenotypes and of missense variants with hypertrophic and restrictive phenotypes. The findings extend these genotype-phenotype patterns to an admixed Latin American population and underscore the importance of improving the representation of diverse populations in genomic reference databases. The observed associations of nonsustained VT, low QRS voltage, and myocardial fibrosis with adverse outcomes further suggest that electrical and tissue-characterization markers may complement LVEF in future risk-stratification strategies. These observations require confirmation in larger, prospectively characterized, and independent cohorts before informing individualized clinical decision-making.

ARTICLE INFORMATION

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Disclosures

None

Supplemental Material

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